

White River stage inundation (Nora): interview note

USGS 03351000 / NWS NORI3. One 5 km drain-to-reach window. Two water surfaces on one HAND grid.

Revisions: 2026-08-28: package the frozen pair (flood stage 11.00 ft and crest 21.18 ft on 2026-08-15, NWS provisional). Generated 2026-08-28T05:24:35Z. Git ae68fbb07f62. v1 PNG sha256 cab5c15439bb.

Result

Flood stage is a tight bathtub (Delta = 1.09 m, 1197 wet of 2604 drain-to-reach). At 21.18 ft (provisional), wet cells are 1876. Extra 679 wet cells filled leftover SFHA (dry 369 to 50); unshaded X wet 38 to 338. IoU 0.73 to 0.76 on drain-to-reach. That is the crest figure's result. Zone X in the Nora window starts to light up at 21.18 ft, still under a 30 m HAND assumption. Not a new model.

WSE is gage zero plus stage in NAVD88: $710.51 + 11.0 = 721.51$ ft, and $710.51 + 21.18 = 731.69$ ft. `h_channel` is the sibling DEM at the White River cell (218.82 m), not gage zero. Paint uses stage and WSE, not discharge.

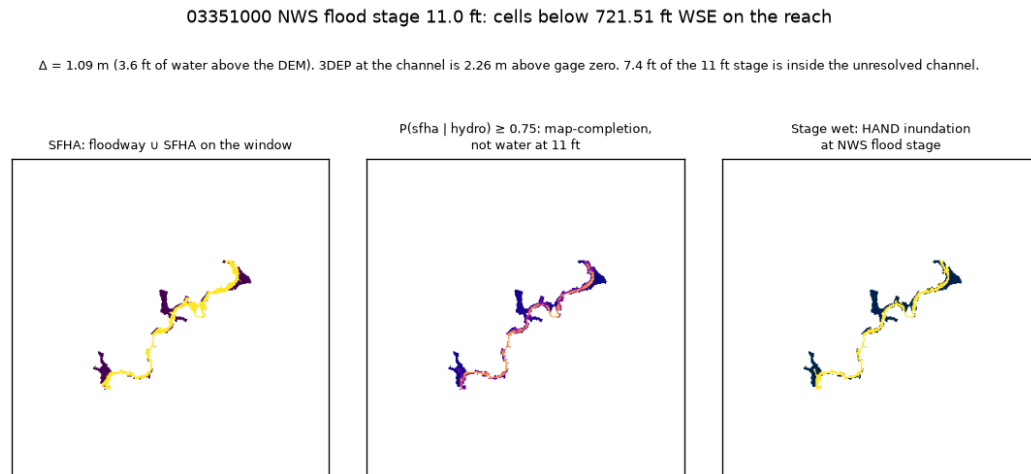
$P(\text{sfha} \mid \text{hydro})$ at $t = 0.75$ is sibling map-completion, not water at 11 ft and not water at 21.18 ft.

Table 1. Same window, two stages (drain-to-reach cells)

Quantity	Flood stage 11.00 ft	Crest 21.18 ft
WSE (ft NAVD88)	721.51	731.69
<code>h_channel</code> (m)	218.82	218.82
Delta (m)	1.09	4.19
Wet cells	1197	1876
Drain-to-reach	2604	2604
SFHA dry at stage	369	50
Unshaded X wet	38	338
IoU SFHA vs wet	0.73	0.76

IoU is on drain-to-reach cells only. Datum is 710.51 ft NAVD88. Crest date 2026-08-15, NWS provisional.

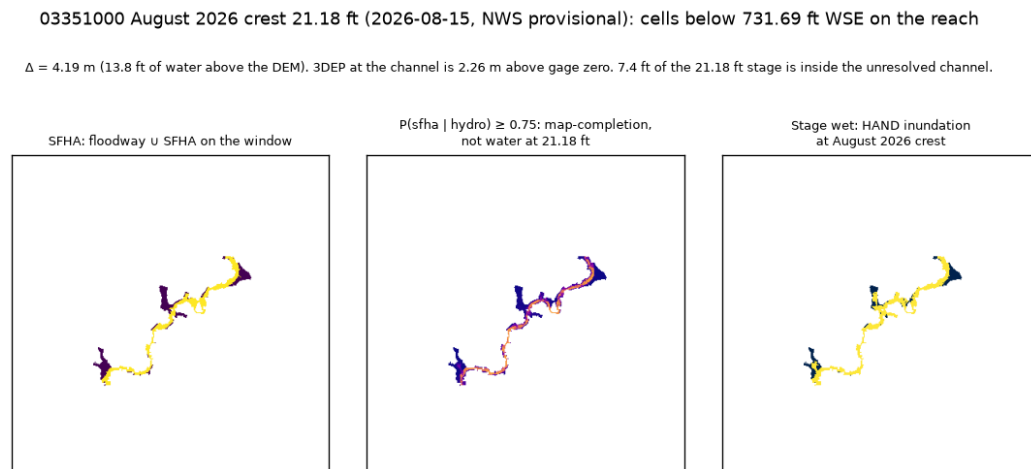
Figure 1. NWS flood stage 11.00 ft



$P(\text{sfha} \mid \text{hydro}) \geq 0.75$ is sibling map-completion, not water at 11 ft. IoU SFHA vs stage wet = 0.73 on drain-to-reach cells only.

Cells below 721.51 ft WSE on the 5 km reach, Delta = 1.09 m (3.6 ft of water above the 30 m DEM), 3DEP at the channel is 2.26 m above gage zero.

Figure 2. August 2026 crest 21.18 ft (2026-08-15, NWS provisional)



$P(\text{sfha} \mid \text{hydro}) \geq 0.75$ is sibling map-completion, not water at 21.18 ft. IoU SFHA vs stage wet = 0.76 on drain-to-reach cells only. Extra 679 wet cells filled leftover SFHA (dry 369 to 50); unshaded X wet 38 to 338. IoU 0.73 to 0.76 on drain-to-reach.

Cells below 731.69 ft WSE on the same reach. Delta = 4.19 m (13.8 ft above the DEM). Extra 679 wet cells filled leftover SFHA (dry 369 to 50); unshaded X wet 38 to 338. IoU 0.73 to 0.76 on drain-to-reach.

Limits

A 30 m HAND model cannot place the unresolved 7.4 ft of stage onto the floodplain. Window D8 is rebuilt from DEM plus HAND=0 paint; it is not byte-identical to sibling Stage B. HAND values were not recomputed. Sibling indiana_flood_completion was not edited. No third Delta, no second gage, no HUC-wide paint.